

Resource Trajectories

1 Main Idea

Resource Trajectory

A **resource trajectory** is a graph that shows the combined progress of two processes and helps visualize safe and unsafe states in deadlock avoidance.

The graphical method is mainly for understanding. It does not directly become a practical operating-system algorithm, but it explains why some resource requests must be delayed.

Connection to Deadlock Avoidance

Deadlock avoidance is based on refusing a resource request if granting it would move the system into an unsafe state.

2 The Two-Process Model

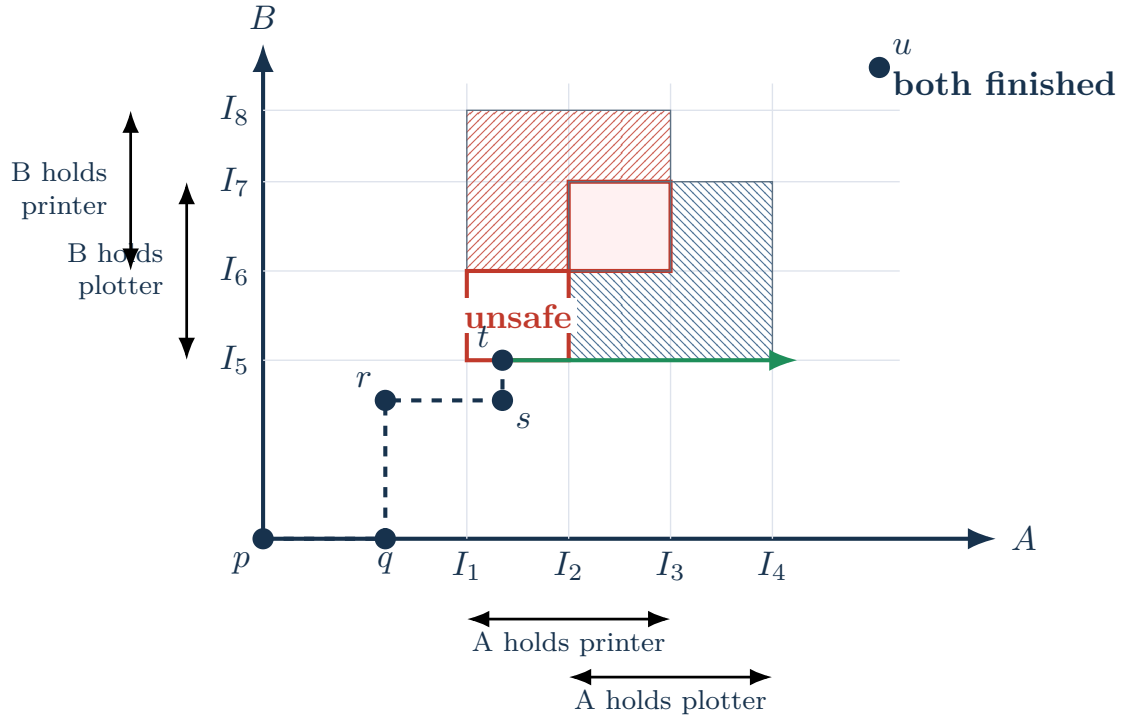
Assume two processes, A and B , and two resources: a printer and a plotter.

- The horizontal axis shows how many instructions process A has executed.
- The vertical axis shows how many instructions process B has executed.
- Each point in the graph is a joint state of both processes.
- On a single processor, movement is horizontal or vertical, not diagonal.

Direction of movement

The path can move only east or north. Processes move forward in time, so the trajectory cannot move west or south.

3 Resource Trajectory Diagram



Marking	Meaning
	Forbidden region: both processes would hold the printer.
	Forbidden region: both processes would hold the plotter.
	Unsafe region: entering it can lead to deadlock.
	Safe choice at t : run A until it releases the plotter.

Figure: Resource trajectory for two processes and two resources.

4 Meaning of the Lines

Point or line	Meaning
p	Initial state: neither process has executed instructions.
q	Process A has run; process B has not run yet.
I_1	A requests the printer.
I_2	A requests the plotter.
I_3	A releases the printer.
I_4	A releases the plotter.
I_5	B requests the plotter.
I_6	B requests the printer.
I_7	B releases the plotter.
I_8	B releases the printer.
u	Both processes have finished.

5 Forbidden Regions

Forbidden Region

A **forbidden region** is a part of the graph that cannot be entered because it would mean both processes hold the same exclusive resource at the same time.

- One shaded region means both processes would have the printer.
- The other shaded region means both processes would have the plotter.
- Mutual exclusion makes both regions impossible.

6 Unsafe Region

Unsafe Region

An **unsafe region** is a state region from which the system may be forced into deadlock if scheduling continues badly.

In the diagram, the unsafe box is bounded by:

$$I_1 < A < I_2 \quad \text{and} \quad I_5 < B < I_6$$

Inside this region:

- A already holds the printer and will soon request the plotter.
- B already holds the plotter and will soon request the printer.
- At the intersection of I_2 and I_6 , both processes wait for each other.

Deadlock point

At $I_2 \cap I_6$, process A asks for the plotter while process B asks for the printer. Each resource is already held by the other process.

7 Scheduler Decision at t

At point t , process B requests the plotter. The operating system must decide whether to grant it.

Decision	Result
Grant B 's request	The system enters the unsafe region and may deadlock later.
Suspend B	A can continue until it releases the plotter, keeping the system safe.

Correct action

At point t , the safe action is to run process A until it passes I_4 , where A has released the plotter.

8 Important Observations

- A safe state is not necessarily deadlock-free forever; it means there exists at least one safe path to completion.
- An unsafe state is not deadlock yet, but it can lead to deadlock.
- Avoidance algorithms prevent entry into unsafe states.
- The trajectory diagram gives intuition for safe states, but real systems need algorithmic checks.